

ACME CORP.

RAG Pipeline Evaluation Matrix: 20 Test Queries & Ground Truth

Evaluation Framework Guide: This suite is engineered to assess retrieval and generation capabilities across 4 categories:

- **Direct Retrieval:** Tests basic semantic match and extraction within a single policy block.
- **Disambiguation:** Tests if the retriever gets confused by overlapping tokens ("leave", "manager", "budget") across different files.
- **Multi-Hop Reasoning:** Requires synthesizing facts scattered across two or more sections to construct a valid answer.
- **Negative Constraints:** Tests if the system correctly admits missing information instead of hallucinating.

ID	TEST CATEGORY	TEST QUERY	EXPECTED GROUND TRUTH & EVALUATION FOCUS
Q01	DIRECT RETRIEVAL	How many days in advance must a leave request be submitted?	Must state at least fourteen (14) days prior via the HR Portal. (Source: Leave Policy 1.1)
Q02	DIRECT RETRIEVAL	What is the standard limit for Client Entertainment expenses and which budget gets hit?	Limit is \$150 per event . It impacts the Marketing / Sales Budget . (Source: Expense Policy Matrix)
Q03	DIRECT RETRIEVAL	What happens if a travel request exceeds the standard baseline cap by more than 20%?	It forces a mandatory route for higher-level approval from the divisional Director . The line manager cannot override it. (Source: Travel Policy 3.1)
Q04	DIRECT RETRIEVAL	How quickly must an employee report a misplaced corporate asset while traveling?	Must file an incident report within 2 hours . (Source: Security Policy 4.2)
Q05	DIRECT RETRIEVAL	What is the exact limit for Hardware / Mobile Device expenses and who must approve it?	Limit is \$500 per item . Requires joint approval from the Manager + IT Security Head . (Source: Expense Policy Matrix)
Q06	DISAMBIGUATION	When taking leave, when is a manager required to review departmental budget allocations?	Only when the leave exceeds ten (10) consecutive business days . (Tests if "leave + budget" targets HR policy rather than Expense/Travel).
Q07	DISAMBIGUATION	Under what specific condition is remote VPN access automatically suspended due to absence?	When marked on extended leave (medical, parental, sabbatical) exceeding 15 business

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			days. (Tests separation of Leave rules from Security rules).
Q08	DISAMBIGUATION	If a manager denies approval for a leave block, can the employee still log the time off?	No. If denied due to operational strain, the employee cannot log the time off. (Isolates Leave policy 1.1).
Q09	DISAMBIGUATION	Whose budget absorbs the cost of replacing equipment misplaced during business travel?	It is not absorbed by IT ; it is charged directly to the responsible employee's departmental operations budget . (Tests "travel + budget + equipment").
Q10	DISAMBIGUATION	What happens if a travel authorization code is missing from an international travel expense submission?	The manager has no authority to grant approval , and the budget allocation will be blocked . (Isolates Expense 2.2 constraint).
Q11	MULTI-HOP	If an employee's pre-approved vacation is canceled for a business emergency, how is the reimbursement handled and whose approval is needed?	Must be routed through the expense workflow , requires VP/Director sign-off (secondary approval) , and is validated against the emergency travel/HR contingency budget . (Combines Leave 1.2 and Expense Matrix).
Q12	MULTI-HOP	What happens if an employee incurs dining or meal expenses while officially registered as being on annual leave?	It is treated as an explicit security breach . The auditing engine flags it and escalates the case to the Information Security team . (Combines Expense 2.2 and Leave 1.2).
Q13	MULTI-HOP	Can an employee extend a corporate business trip for a personal vacation, and what are the prerequisites?	Yes, if personal leave duration ≤ business duration . Prerequisites: Must obtain separate leave approval from their manager prior to booking and pay out-of-pocket. (Combines Travel 3.2 and Leave 1.1).
Q14	MULTI-HOP	What cross-system data does ACME continuously monitor to catch malicious insider behavior or credential theft?	Algorithms compare travel itineraries, approved expense receipts, leave calendars, and active network login logs . (Synthesizes checks across Travel, Expense, Leave, and Security).
Q15	MULTI-HOP	If corporate funds are accidentally used to extend lodging during a personal vacation attached to a business trip, what happens?	It violates the corporate asset policy, triggers an internal security review , and results in the automatic revocation of corporate card privileges . (Combines Travel 3.2 and Expense 2.2).
Q16	NEGATIVE CONSTRAINT	What is the exact dollar limit allowance for hotel	Not stated / Refuse to answer. The text mentions "baseline caps" but never specifies a

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		accommodations per night under the Travel Policy?	night-by-night dollar limit. (Tests baseline hallucination protection).
Q17	NEGATIVE CONSTRAINT	Does an employee get paid out for unused leave balance at the end of the year?	Not stated / Out of scope. The handbook does not discuss rollover, expiration, or payout rules. (Tests context boundary containment).
Q18	NEGATIVE CONSTRAINT	What is the password complexity requirement for ACME corporate accounts?	Not stated. While access control is detailed in Section 4.1, specific credential/password syntax requirements are absent. (Tests context boundaries).
Q19	NEGATIVE CONSTRAINT	What specific flight booking tier (e.g., Economy, Business Class) is permitted for international flights?	Not stated. The text covers flight authorizations and budget caps, but omits permissible ticket classes. (Checks if the system guesses "Economy").
Q20	NEGATIVE CONSTRAINT	How long must a line manager serve at ACME Corp before they are allowed to approve team budgets?	Not stated. Tenure requirements for manager authorization privileges are not provided in the handbook text. (Tests restraint against logical assumptions).